TEXORA · SESSIONAL NOTES · STUDY SHEET

Pattern Making for Basic Trouser

Drafting the front and back trouser block from the TE-3108 chart: construction frame, quarter-girth dispositions, fork curves, the back seat angle and dart, and the walking checks that validate the draft.

TE-3108 · APPAREL MANUFACTURING TECHNOLOGY-II SESSIONAL · STUDY 3 · LEVEL 3 TERM 1

1. Aim and place of the study

Aim: to draft the basic trouser block — front and back — from the TE-3108 measurement chart (size M), deriving every construction value from the chart and recording each derivation on the draft.

2. Materials, machine settings and specimens

Materials: pattern paper, L-square, long ruler, tape, French curve, pencil, notcher and awl. **Working size M:** waist 82, hip 102, outseam 104, inseam 78, thigh 62, knee 46, hem 40, front rise 26. From the chart: rise depth check — outseam − inseam = 104 − 78 = 26 cm, which must equal the body rise including the waistband; the waistband is 4 cm wide, drafted separately. Quarter values used in drafting: quarter waist = 20.5, quarter hip = 25.5.

Trouser measurement chart used throughout TE-3108 (men's basic trouser, garment values in cm): waist S 77 · **M 82** · L 87 · XL 92 (grade 5); hip 97 · **102** · 107 · 112 (grade 5); outseam (incl. waistband) 102 · **104** · 106 · 107; inseam 76 · **78** · 80 · 81; thigh 60 · **62** · 64 · 66 (grade 2); knee 44 · **46** · 48 · 50 (grade 2); hem (bottom) width 38 · **40** · 42 · 44 (grade 2); front rise 25 · **26** · 27 · 28. Size M is the working size unless the study states otherwise. Seam allowances, as in TE-2207/TE-2208: **1 cm** general, **1.5 cm** at side seams, waistband and other adjustment seams, **2.5 cm** at hems. The shirt values of TE-2207/2208 (size M chest 107, length 73, shoulder 46, sleeve 61.5, collar 40; grades 5/2/2/1.5/2 cm) are used unchanged wherever a shirt is the specimen.

3. Drafting the front

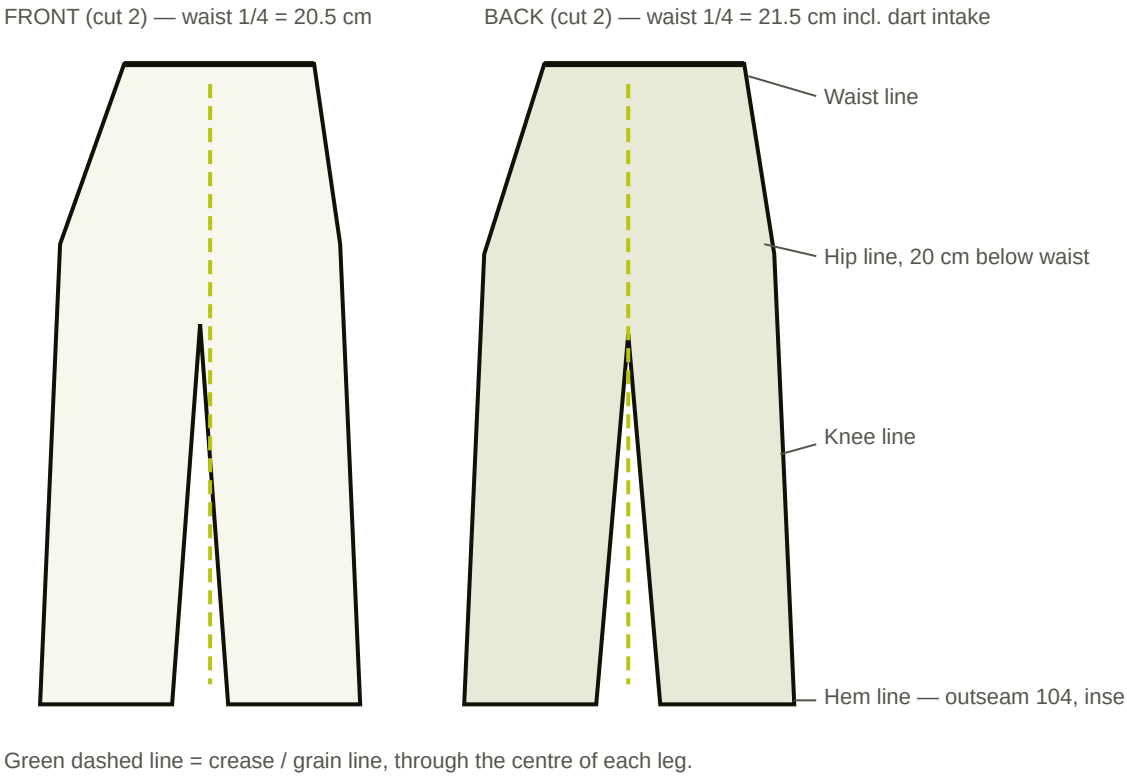


Figure — Basic trouser draft, front and back, size M of the chart in §3. Lengths full; widths drafted as quarter girths.

1. Draw the crease line vertically. From the top (waist line) mark down: hip line at 20 cm (waist-to-hip depth of this system), crutch line at the rise depth 26 cm, knee line midway between crutch and hem (adjusted per system), hem line at outseam 104 cm from the waist (inseam 78 from the crutch).
2. On the hip line set quarter hip 25.5 cm disposed about the crease line (front hip width measured from the side seam to the front crutch extension). Extend the front crutch point by about 4 cm beyond the hip width line for the fork curve.
3. On the waist line set quarter waist 20.5 cm plus a pleat/dart allowance as the style requires (flat front here: no pleat; the small excess is eased). Shape the fly curve from the waist at centre front down to the crutch point with the French curve.
4. Thigh 62 → quarter 15.5 each side disposition about the crease at crutch level; knee 46 and hem 40 are set symmetrically about the crease line (half values 23 and 20 disposed equally), so the crease stays central as the leg tapers.
5. Join side seam (waist–hip–knee–hem, gently curved) and inseam (crutch point–knee–hem). Check: inseam measured on the draft = 78 cm.

4. Drafting the back

1. Draft on the front's frame. Back quarter waist = 20.5 cm **plus a waist dart intake of 2 cm** taken inside the piece, so the back waist seam length after the dart is sewn equals the front's disposition on

the waistband.

2. Raise the back waist at centre back by about 2 cm (the seat angle) and extend the back crutch point about 5 cm with a deeper fork curve — the back rise seam measures longer than the front's, as sitting requires.
3. Back hip width uses the same quarter hip 25.5 with its disposition shifted toward the side seam; back knee and hem take the same widths as the front at this basic stage, distributed about the same crease positions.
4. Draw the waist dart: centre at the midpoint of the back waist, 2 cm intake, length about 10 cm, ending in a drill hole 1 cm before its point.

5. Checking the draft

- Rise check: outseam − inseam = 26 on both pieces.
- Walk front and back inseams and outseams against each other — equal lengths (outseam ease of about 0.5 cm in the back is acceptable and recorded).
- Waist check: front quarter + back quarter (after dart) = 20.5 + 20.5 = 41 = half of waist 82 — the full waistband length check of Study 5 depends on this.
- Crease lines central at knee and hem on both pieces.

Precautions

- Derivations are written on the draft as they are used; an unlabelled number cannot be examined or corrected.
- Add no seam allowance in this study — allowances belong to the production-pattern stage and are applied piece by piece in Studies 4 and 5.
- Walk the seams with the tape on edge before inking the final lines.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Verify the rise from the chart values, and state what the 4 cm waistband does to the outseam measure as drafted.
2. Calculate the quarter waist and quarter hip of size L from the chart.
3. Why is the back crutch extension longer and its fork curve deeper than the front's?
4. The drafted inseam measures 77.2 cm against a chart value of 78. Where do you look first?

Answer key

1. Outseam 104 − inseam 78 = 26 cm of rise including the waistband; as drafted to the top of the waistband the outseam is the full 104, so the leg draft itself carries rise 26 − 4 = 22 cm below the band's lower edge — the point at which the band is attached in Study 5.
2. Size L: waist 87 ÷ 4 = 21.75 cm; hip 107 ÷ 4 = 26.75 cm.
3. The back must accommodate the seat's roundness and the sitting posture: more length and a deeper curve over the buttocks, or the trouser pulls down at the back waist and strains at the fork.
4. At the vertical positions: crutch-line depth (rise) and hem-line placement. A length error on a straight run is almost always a misplaced horizontal line, not a curve error — re-check the two depths against the frame before redrawing curves.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib